

HAT-P-54b

R-1758

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_171116 · created 2026-08-02T20:39:34+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458073.8355 +/- 0.00022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1999 +/- 0.005
Transit depth (Rp/Rs)^2	4.0 +/- 0.2 [%]
Orbital Inclination (inc)	89.69 +/- 0.65
Airmass coefficient 1 (a1)	1.08 +/- 0.046
Airmass coefficient 2 (a2)	-0.078 +/- 0.012
Scatter in the residuals of the lightcurve fit is	4.22 %
Best Comparison Star	None
Optimal Aperture	3.11
Optimal Annulus	4.83
Transit Duration (day)	0.07605 +/- 0.0005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1999 $\pm$ 0.005 vs 0.1572 (deviation 5.86 σ)
Duration fit vs published	0.07605 d vs 0.075 d (deviation 1.44 σ)
Mid-time O-C	-1.2 min
Depth SNR	27.4
Residual scatter	4.22 %

Light curve

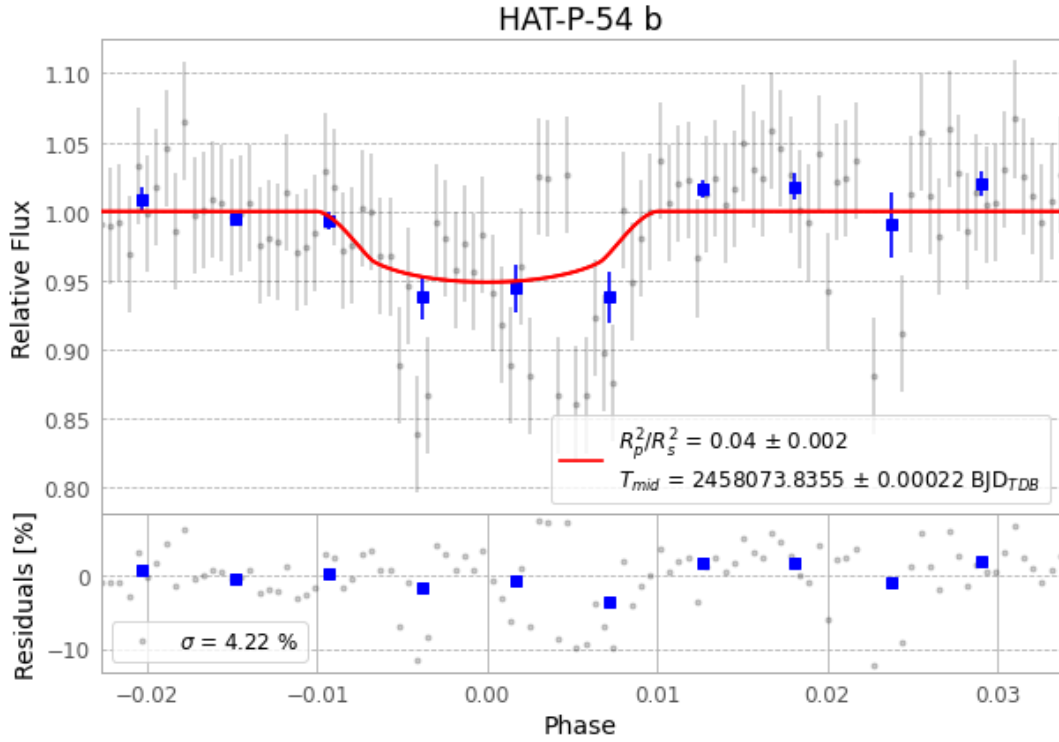


Figure 1 — FinalLightCurve_HAT-P-54 b_2017-11-15.png (SHA-256 dd3bb7dab9e9eb43...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 203], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6a4650d924d33f56...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

104 FITS frames (68 MB), HATP-54171116055112.FITS → HATP-54171116110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-171116.log (4d6eca9939e0...), FinalLightCurve_HAT-P-54 b_2017-11-15.png (dd3bb7dab9e9...), FinalLightCurve_HAT-P-54 b_2017-11-15.csv (630e64fdd2e8...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 20:39Z.